

AWS EFS

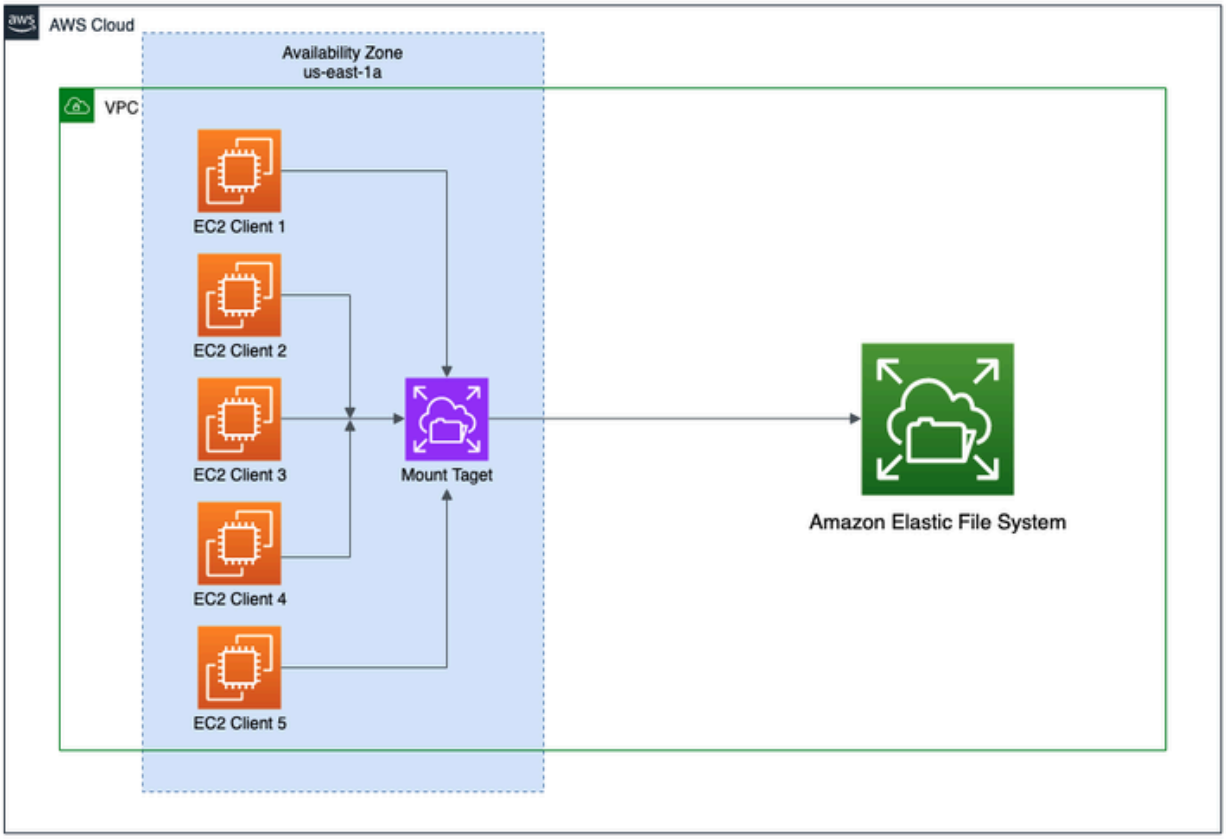
Amazon Elastic File System

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About EFS:

Amazon Elastic File System (Amazon EFS) provides a simple, scalable, fully managed elastic NFS file system for use with AWS Cloud services and on-premises resources. It is built to scale on demand to petabytes without disrupting applications, growing and shrinking automatically as you add and remove files, eliminating the need to provision and manage capacity to accommodate growth. Amazon EFS is designed to provide massively parallel shared access to thousands of Amazon EC2 instances, enabling your applications to achieve high levels of aggregate throughput and IOPS with consistent low latencies.



EFS Features & Use Case:

Features:

- Fully Managed: AWS handles storage management, patching, scaling, and availability.
- Elastic Scaling: Grows and shrinks automatically as files are added/removed.
- High Availability: Data is stored across multiple Availability Zones (AZs) in a region.
- Multiple EC2 Mounts: Many instances can read/write to the same file system.
- POSIX-Compliant: Works like a standard Linux file system with file locking, permissions, etc.
- Pay as you go: You pay for the storage you actually use.

Amazon EFS does not natively support mounting from Windows EC2 instances. For Windows, you can choose FSx.

Use Case:

- Web Server Content Sharing
- CMS Platforms
- Big Data & Analytics
- Development Environments
- and many more!

Performance Modes:

General Purpose
Default mode for most applications.
Suitable for latency-sensitive applications.

Max I/O
Higher throughput.
Slightly more latency, better for thousands of connections.

EFS Storage Classes:

EFS Standard
General-purpose, highly durable and available.
Best for frequently accessed files.

EFS Infrequent Access (IA)
Lower cost for files accessed less often.
Slightly higher retrieval latency than Standard.

EFS One Zone
Stores data in a single Availability Zone.
Cheaper than multi-AZ storage, but lower redundancy.

EFS Lifecycle Management
Automatically moves files from Standard, IA based on last access.
Helps optimize cost without manual intervention.

EFS Availability and Durability:

Standard/Regional:
Multi AZ
Great for Production

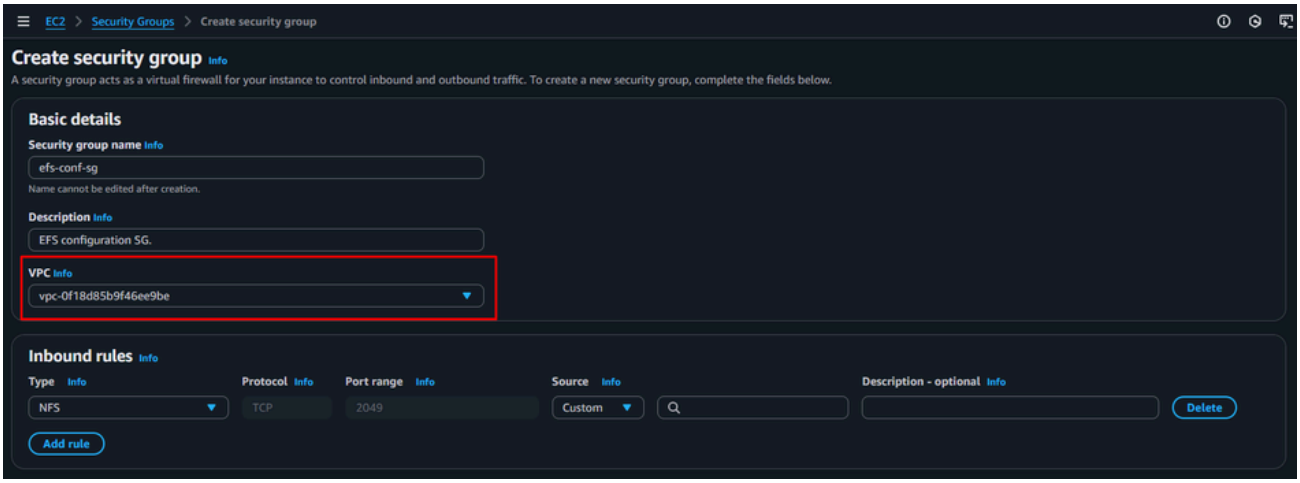
One zone:
Resides in only 1 AZ
Great for Development
If used with EFS-IA == more discount ,up to 90%

Configure EFS:

Before configuring EFS, let's first create a Security Group (SG) that will help us connect to multiple EC2 instances in same network.

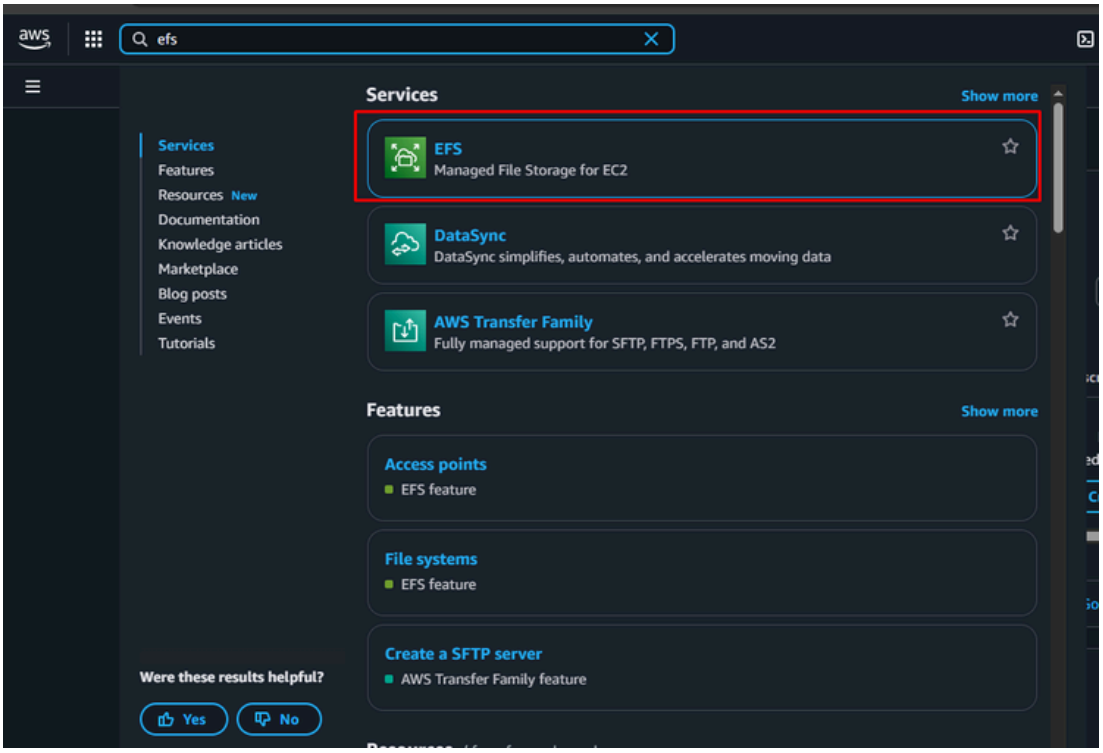
You must select a subnet before you can add an EFS file system.

To configure a new SG, log in to your AWS account and search for "Security Group." Then configure your SG. If you have an existing one, you can use it—there's no need to create a new one. In my case, I created a new one. You must choose a VPC for your SG.



Configure EFS:

After creating SG, now search for EFS. Then click on EFS...

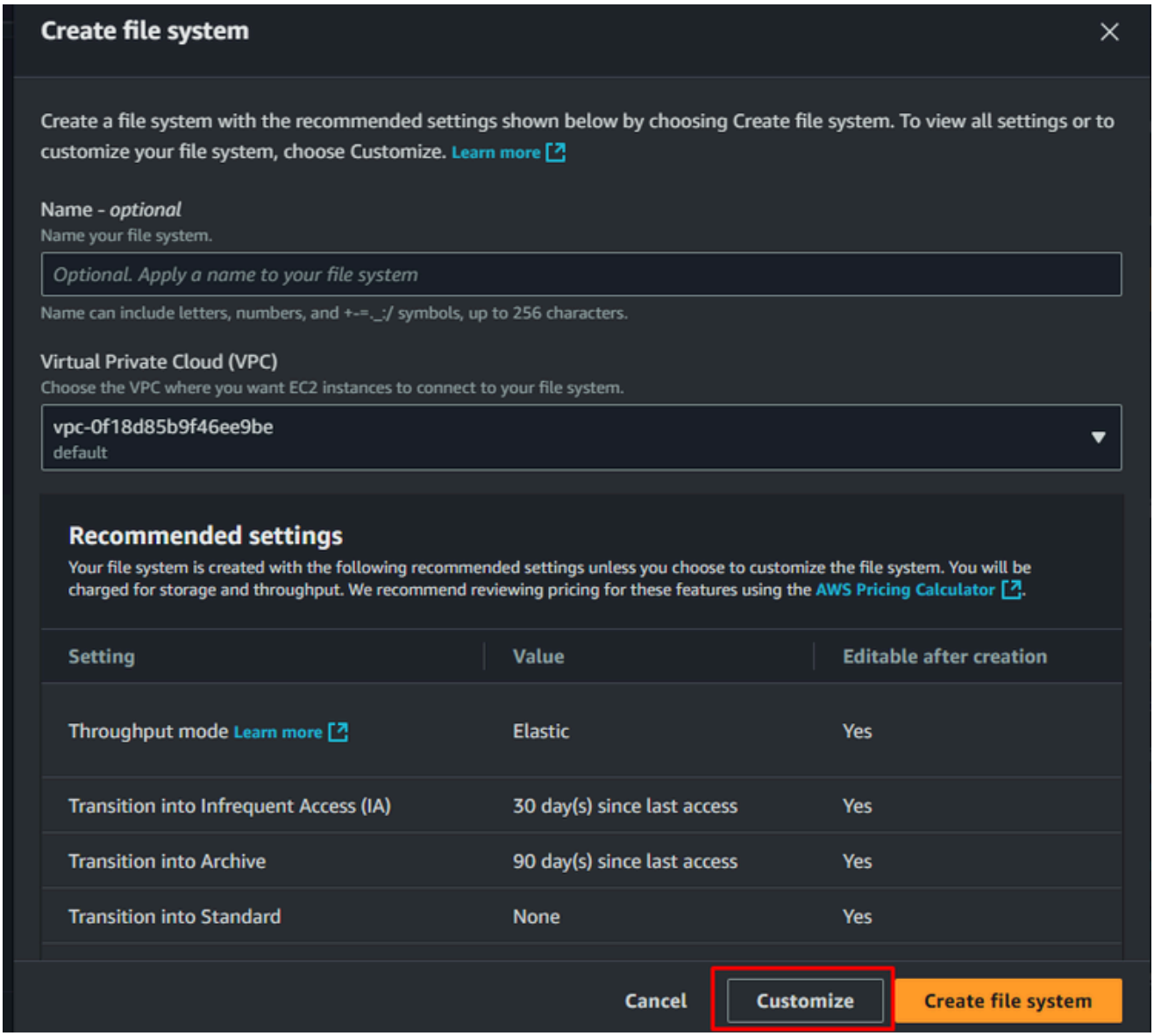


Now, you get this page. Click on “Create File System” button...



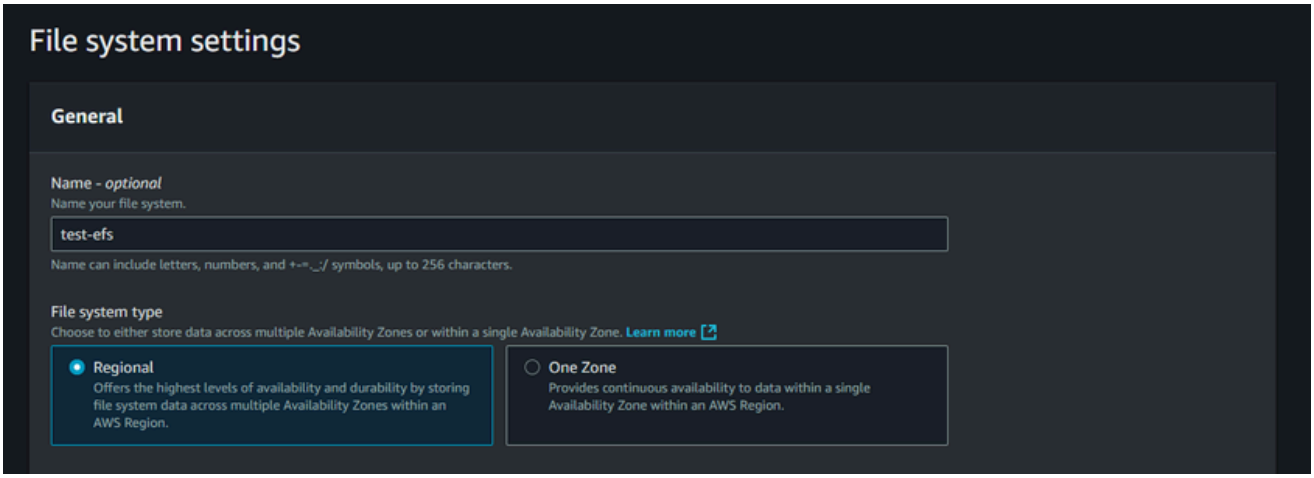
Configure EFS:

After click on Create file system button get this popup, you can naming it and click Create File System or Customize in my case i use customize for more details.

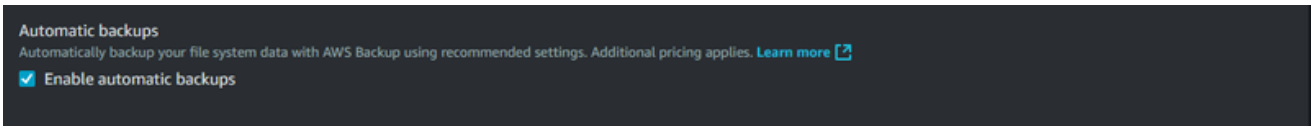


Configure EFS:

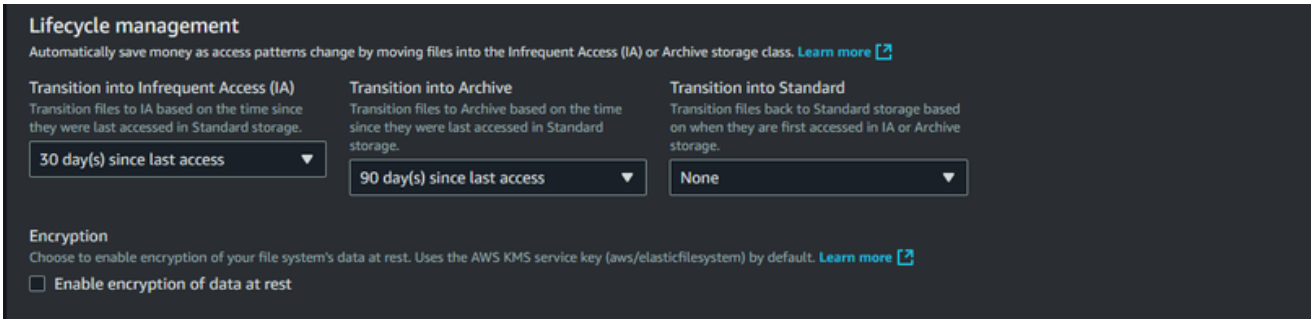
After clicking the Customize button, you will see this page. In the General tab, you get two options: “Regional,” which allows access to all Availability Zones (AZs) in your chosen region, and “One Zone,” which allows access to only the single zone you select.



Then you get another option for backup, If you need backup you select it otherwise ignore.

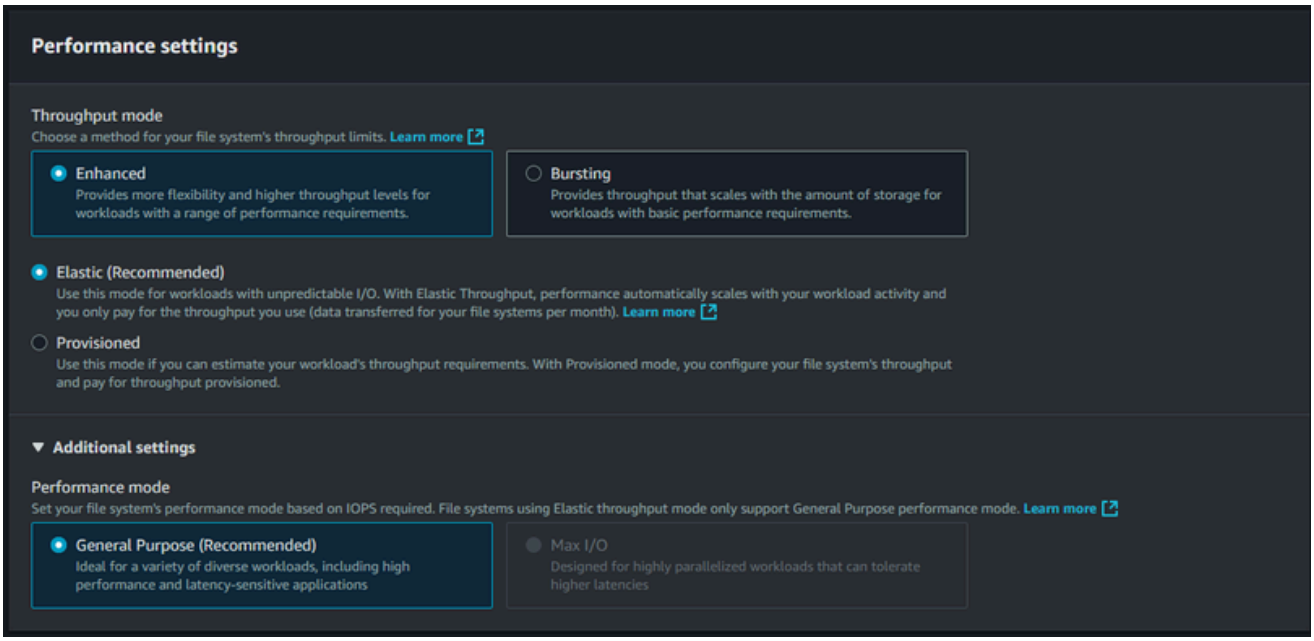


Then you get another option for Lifecycle management and encryption. Just choose those based on your needs.



Configure EFS:

In the Performance tab, you get more options. I kept the default settings because I won't be using it for long. You can configure it based on your needs.



In Tag tab, you can use tag for your NFS.

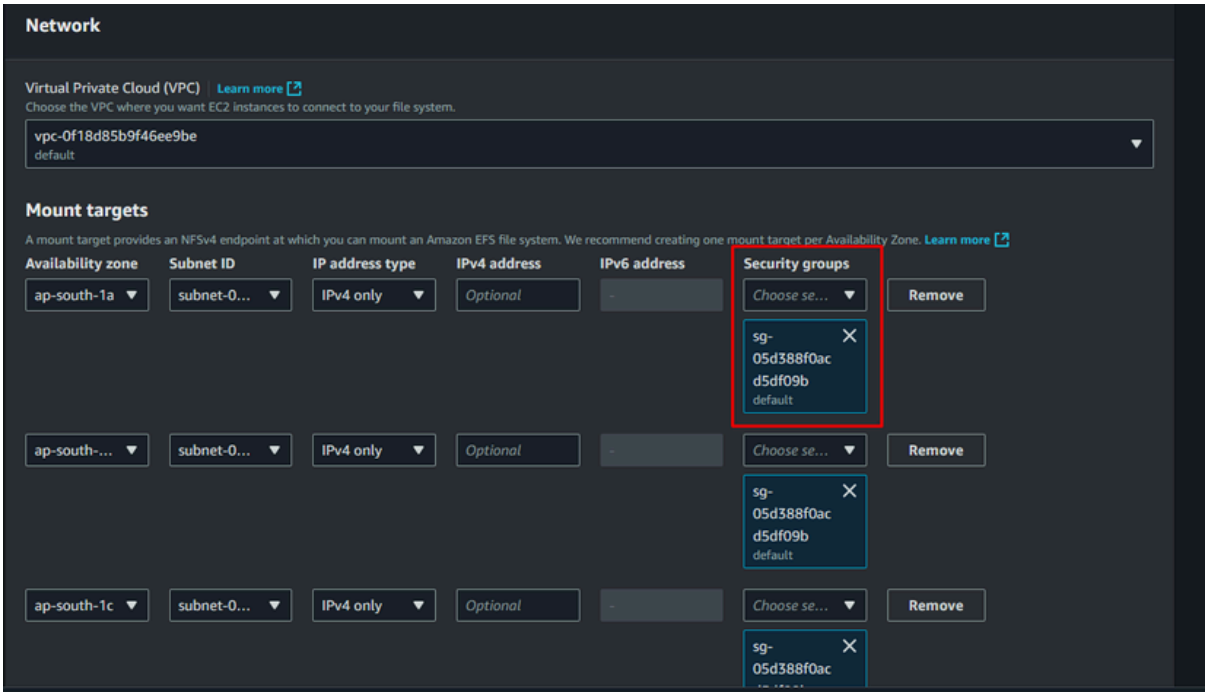


After setup first step click on next button for next step which is network related.

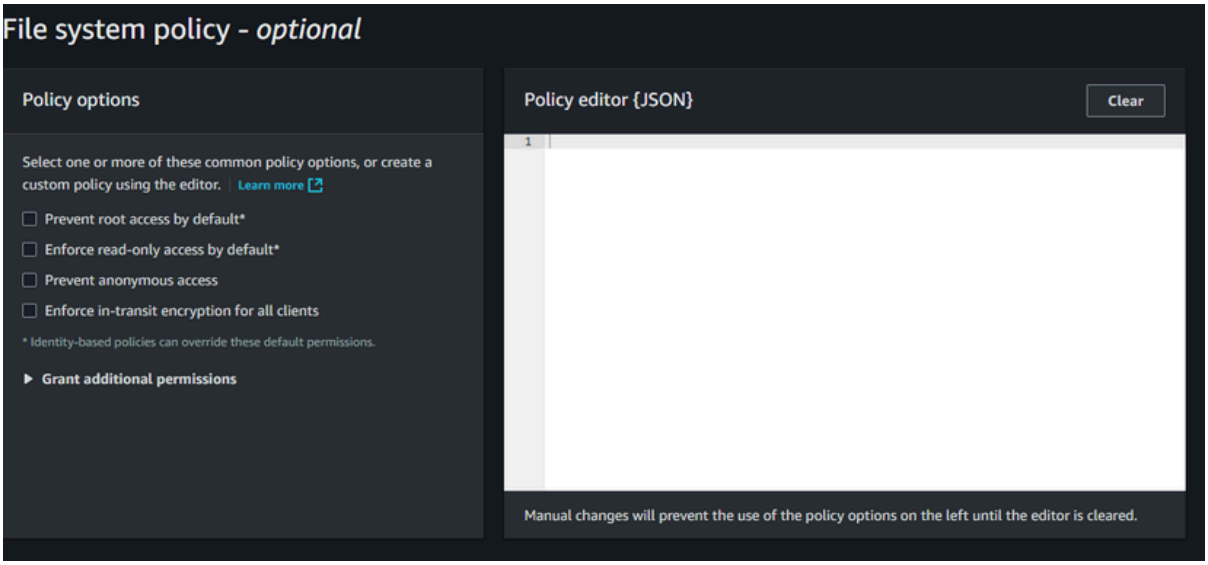


Configure EFS:

In networking step, select SG for your AZ.

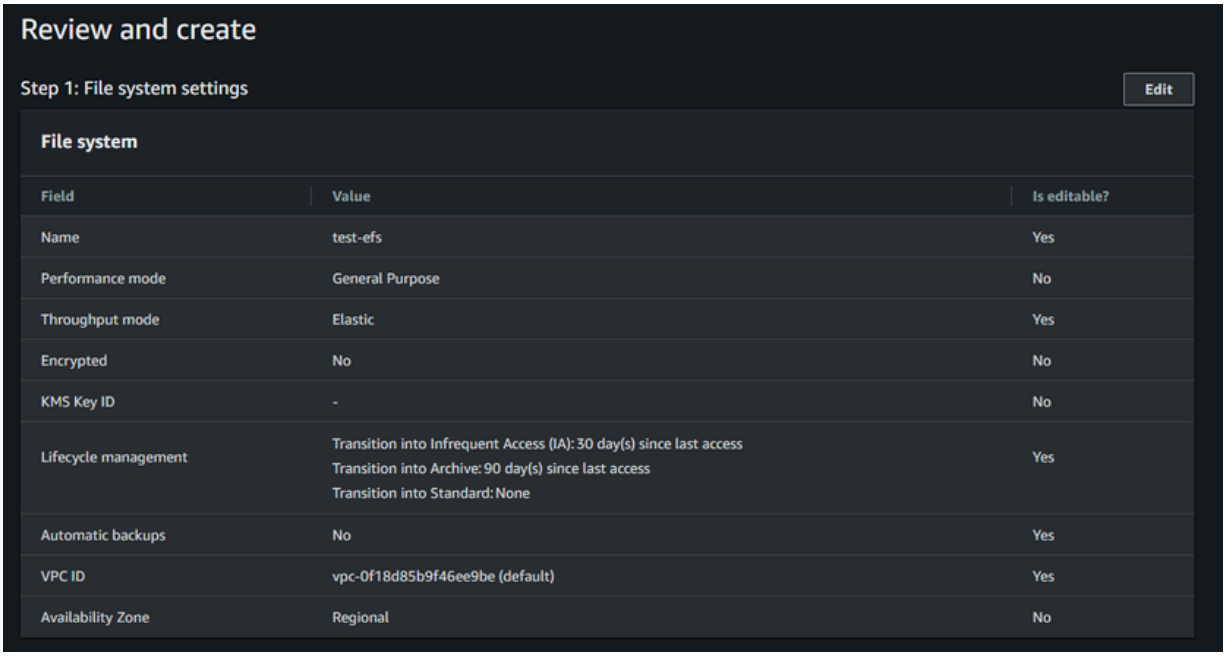


In file policy tab, You can add policy if you need.

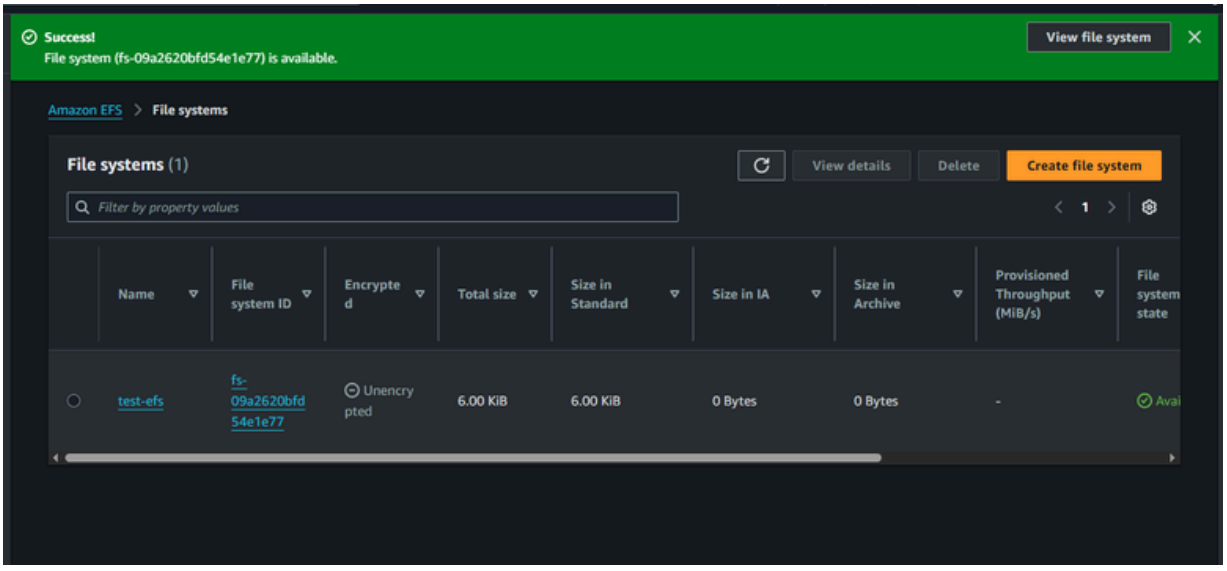


Configure EFS:

Final step, check your configuration and create.



Successfully Created our EFS.



In my project section, I will try to provide a hands-on guide on how to mount EFS on multiple EC2 instances.

Thank You

Stay Connect:

/in/alamgirweb11

/alamgirweb11